

Phase Trajectories of Nonlinear Systems

Augmented Attractor Shaping for Nonlinear Feedback Systems

A Study Project (PHY F266) Report Under
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by

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Introduction

This report outlines control architecture designed to optimize transient trajectories in highly chaotic nonlinear dynamical systems. Based on an assessment of current academic literature across reinforcement learning (RL), nonlinear dynamics, and control theory, the architecture proposed: Residual RL acting on a PID baseline, using Generalized Alignment Indices (GALI) explicitly as state inputs or offline atlases for transient trajectory optimization, represents a significant gap in the literature.

Traditionally, RL and PID are combined in robotics to handle unmodeled friction or noise. Simultaneously, chaos control leverages the Maximal Lyapunov Exponent (MLE) primarily as a reward signal (to suppress chaos). However, using a topological atlas of chaos to feed a Residual RL agent non-Markovian data specifically to optimize the transient energy of a PID-stabilized limit cycle is largely uncharted. By modeling the chaotic bounds and integrating them into real-time RL augmented controllers via sub-millisecond neural surrogates, reductions in the mean as well as the spread of both actuator effort and tracking error are noticed.

The "non-Markovian" part here is critical. The Markov property is required by standard reinforcement learning algorithms. It is assumed under this paradigm (markovity) that the instantaneous state contains all information necessary to predict future dynamics. However in highly chaotic environments no insight into manifold dynamics over time is provided by a single coordinate point. Because the Generalized Alignment Index is mathematically derived by integrating deviation vectors over a finite transient time horizon a direct mapping from the physical state space to a scalar value is created as $f : \mathbb{R}^6 \rightarrow \mathbb{R}$ where the output is $\log_{10}(\text{GALI})$. It is therefore classified as a non-Markovian metric. This paradigm gap is bridged by distilling the temporal metrics into a spatial neural surrogate. The transient evolution of the manifolds is converted into an instantaneous spatial representation. When this mapping is appended to the observation space the non-Markovian nature of the environment is circumvented. The temporal structure is embedded directly into the instantaneous observation which satisfies the standard input constraints of the reinforcement learning algorithm while the required topological foresight for transient trajectory optimization is still provided.

1 Theoretical Foundation

1.1 The 2-Coupled Lorenz System

The benchmark physical model consists of two identical Lorenz oscillators mutually coupled via their x variables. The 6-dimensional state space is defined as $\mathbf{X} = [x_1, y_1, z_1, x_2, y_2, z_2]^T$, governed by the following nonlinear differential equations:

$$\dot{x}_1 = \sigma(y_1 - x_1) + k(x_2 - x_1) \quad (1)$$

$$\dot{y}_1 = x_1(\rho - z_1) - y_1 + u_1 \quad (2)$$

$$\dot{z}_1 = x_1 y_1 - \beta z_1 + k(z_2 - z_1) \quad (3)$$

$$\dot{x}_2 = \sigma(y_2 - x_2) + k(x_1 - x_2) \quad (4)$$

$$\dot{y}_2 = x_2(\rho - z_2) - y_2 + u_2 \quad (5)$$

$$\dot{z}_2 = x_2 y_2 - \beta z_2 + k(z_1 - z_2) \quad (6)$$

The physical constants are the standard chaotic parameters: $\sigma = 10.0, \rho = 28.0, \beta = 8/3$. The coupling strength $k = 2.5$ induces global chaotic synchronization. Control inputs u_1, u_2 are applied to the y -dimensions, with strict actuator saturation limits ($U_{max} = 250$). This system was chosen since a system that had some level of chaos to it was needed, and at the same time, starting off with a hyperchaotic system such as Coupled Rössler network, a 4D hyperchaotic Chen system, or a time-delay Mackey-Glass oscillator seemed ambitious.

1.2 Control Objective

The primary objective of the control architecture is the stabilization of the 6-dimensional state vector $\mathbf{X}$ to the origin, defined as the point $\mathbf{X}_{goal} = [0, 0, 0, 0, 0, 0]^T$. The performance of the controller is evaluated (compared to PID baseline) based on its ability to minimize the transient tracking error $\|\mathbf{X} - \mathbf{X}_{goal}\|$ while reducing the total control effort $\int(|u_1| + |u_2|)dt$ under randomized initial conditions.

1.3 Chaos Quantification: GALI vs. MLE

The Maximal Lyapunov Exponent (MLE) only measures chaos in one direction at a time. Because of this, it fails to provide the full structural map of the stability boundaries that our RL agent needs to navigate the system. The Generalized Alignment Index (GALI) of order k monitors the evolution of a k -dimensional volume element. For k unit deviation vectors $\mathbf{w}_1, \dots, \mathbf{w}_k$, the index $GALI_k$ is defined by the norm of their wedge product:

$$GALI_k(t) = \|\mathbf{w}_1(t) \wedge \mathbf{w}_2(t) \wedge \dots \wedge \mathbf{w}_k(t)\| \quad (7)$$

...which in other words is the volume of a k -dimensional parallelepiped formed by the deviation vectors with orthonormal random initialization using QR decomposition.

If the orbit is chaotic, all vectors align with the MLE direction, causing the volume to collapse as $e^{-(\lambda_1 - \lambda_k)t}$. If $k \leq N$ on a stable torus, GALI fluctuates around a non-zero constant. In this architecture, we utilize $GALI_2$ (equivalent to the Smaller Alignment Index, SALI) to detect the primary transition boundary between chaotic alignment and stable vector independence.

Additionally while integrating the 18-dimensional variational equations for $GALI_2$ carries a higher per-step computational cost than the 12-dimensional system required for the Maximal Lyapunov Exponent (MLE), $GALI_2$ is fundamentally more efficient for transient trajectory optimization. MLE requires asymptotic time horizons ($t \rightarrow \infty$) to average out local fluctuations and converge on a reliable stability metric. In contrast, the $GALI_2$ index decays exponentially fast in chaotic regimes, allowing for definitive topological mapping over extremely short (e.g., 10-second) transient windows.

2 Dataset Generation

2.1 The 18-Dimensional Variational System

To extract the stability manifold, exact GALI value computation across the phase space is required. This involves simulating the variational equations to evolve two deviation vectors (for $GALI_2$) alongside the 6D physical state. The combined state vector is $[\mathbf{x}, \mathbf{w}_1, \mathbf{w}_2]$,

evolving via $\dot{\mathbf{w}}_k = \mathbf{J}\mathbf{w}_k$, where $\mathbf{J}$ is the 6×6 instantaneous Jacobian. This 18-dimensional integration using standard numerical methods (e.g., Runge-Kutta 45 via SciPy) is a computational bottleneck, taking $\sim 72\text{ms}$ per 10 second track of system evolution, strictly violating the latency constraints required for 500 Hz (changed later) closed-loop RL control.

2.2 Strategy Multi-Resolution Adaptive Sampling

To map the transition regions of chaos (regions where $0.1 < \text{GALI} < 0.4$), a 3-pass pipeline was utilized:

1. **Multi-Resolution Survey:** Latin Hypercube Sampling (LHS) with an 80/20 Core/Shell split to map dense attractors and divergence boundaries.
2. **SSM (Subspectral Manifold) Boundary Refinement:** Targeted points at the transition edge, and sampled points around it, by applying local Gaussian jitter ($\sigma = 1.2$) to map the fine-grained chaos transition.
3. **Uncertainty-Based Refinement:** Resampling the top 10% error regions of an initial surrogate to map unvisited phase-space geometries.

Multiple batches of the dataset were generated, each differentiable with its sample size. Each dataset batch was tested with each iteration of the GALI surrogates. Among about ten, here are the three used sets:

- **Small:** 2,500 samples.
- **Medium** (27,500 samples): 25,000 samples + 2,500 refinement samples at transition edge.
- **Large** (325,000 samples): 250,000 samples + 75,000 refinement samples at transition edge.

3 GALI Surrogate/Atlas

To resolve the integration bottleneck, successive PyTorch neural surrogates were developed that could be classified into 4 broad iterations. The goal was to map the 6D phase space directly to $\log_{10}(\text{GALI})$. The best performing models of each version are shown in the following sections.

3.1 v1 & v2: Standard MLPs and Mode Collapse

The v1 surrogate (a standard Feed-Forward MLP) achieved an 85x latency speedup (0.84ms vs 72.36ms). However, it suffered from severe mode collapse. Standard Mean Squared Error (MSE) treats all samples equally. Because chaotic regimes dominate the volumetric phase space, the network memorized deep chaos while ignoring the thin, complex transition boundaries crucial for stabilization. v2 attempted to resolve this by drastically increasing the dataset to 25,000 adaptively sampled points, yet standard MSE still allowed the chaotic bulk to overwhelm gradient updates.

3.2 v3: Physics-Informed Weighted Gradients

To force boundary resolution, v3 introduced Residual Blocks to prevent vanishing gradients across the steep exponential cliffs of Lyapunov exponents. The MSE loss was manually weighted based on physical zones:

- Deep Chaos ($\text{GALI} < 0.1$): Weight = 1.0
- Stable Regime ($\text{GALI} > 0.4$): Weight = 3.75
- Transition Zone ($0.1 < \text{GALI} < 0.4$): Weight = 5.0

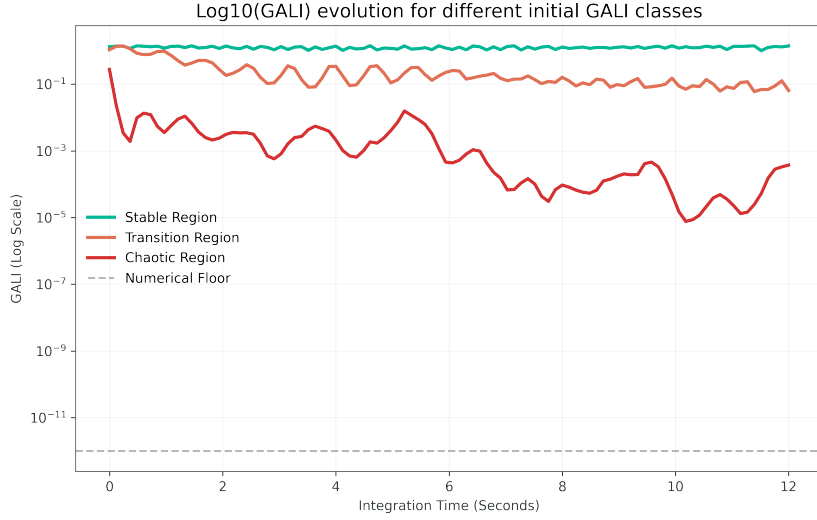


Figure 1: Temporal Evolution of $\log_{10}(\text{GALI})$ Across Dynamical Regimes. Representative trajectories illustrating stable (green), transition (orange), and chaotic (red) regimes. Stable trajectories maintain high GALI values, chaotic trajectories exhibit rapid exponential decay toward numerical floor, and transition trajectories display intermediate, fluctuating behavior. This separation justifies the definition of distinct dynamical classes used in surrogate training and adaptive sampling.

These numbers were taken based on initial GALI value track behaviour.

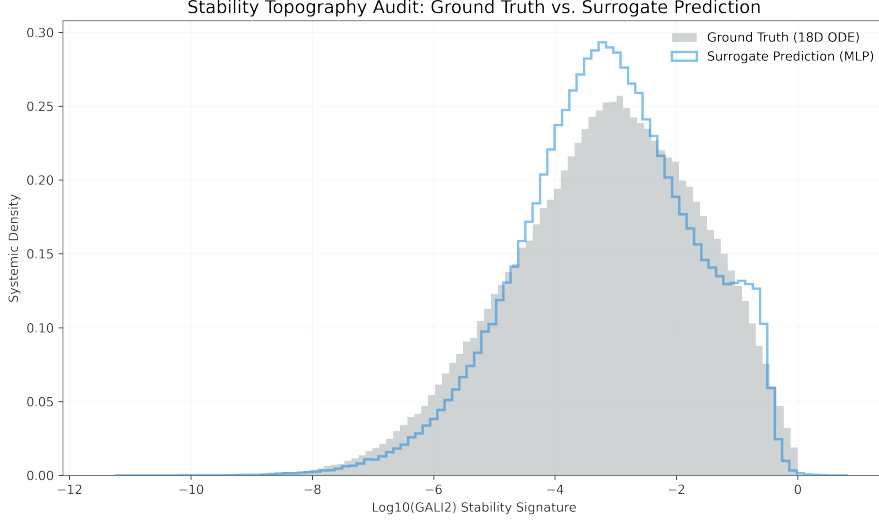


Figure 2: V3 Surrogate Prediction Audit. Histogram comparison between ground-truth $\log_{10}(\text{GALI})$ values and V3 model predictions.

Despite physics-informed weighting, the model exhibits bias toward high-density chaotic regions, with under-representation of the transition manifold, even after iterations of adjustment. The presence of mismatch indicates incomplete resolution of thin stability boundaries. Hard-coded thresholds created artificial ridges in the continuous output (GALI_2) space. The RL agent later exploited these ridges as false local minima, leading to inaccurate residual corrections that worsened the overall augmented controller.

3.3 v4: Inverse-Density Distribution Matching (Final)

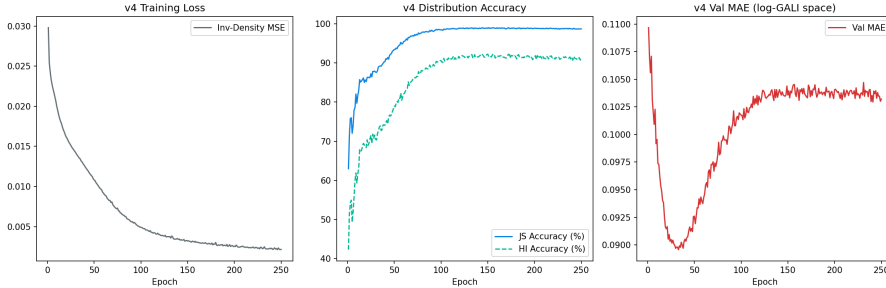


Figure 3: V4 Training Audit. Training and validation loss curves over epochs, along with Jensen-Shannon (JS) divergence tracking.

Manual thresholds were removed and implemented a data-driven, distribution-aware loss function. The GALI dataset was mapped into 200 bins, and each sample was assigned a loss multiplier equal to $1/\sqrt{\text{count}}$. This inverse-density weighting naturally down-weighted the dense chaotic regions and prioritized the rare, highly-folded transition boundaries without breaking manifold smoothness. Convergence demonstrates stable optimization under inverse-density weighting, with JS divergence approaching over a 99 percent distributional alignment.

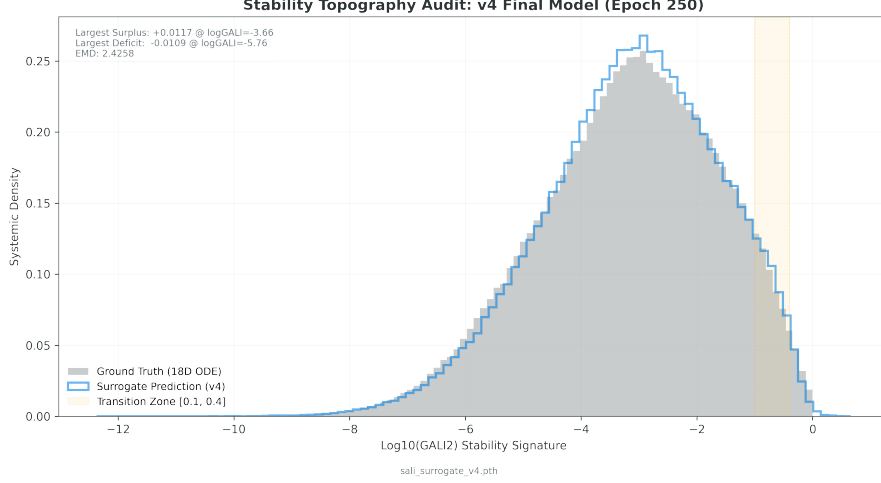


Figure 4: V4 Surrogate Prediction Audit (Inverse-Density Weighted). Distributional alignment between predicted and ground-truth $\log_{10}(\text{GALI})$ values.

Inverse-density weighting significantly improved coverage of low-measure transition regions while still distilling $\log_{10}(\text{GALI})$ values in chaotic regimes. The near-overlap demonstrates successful mitigation of mode collapse and improved manifold continuity. The weight profile used in the final frozen loss function is given as below.

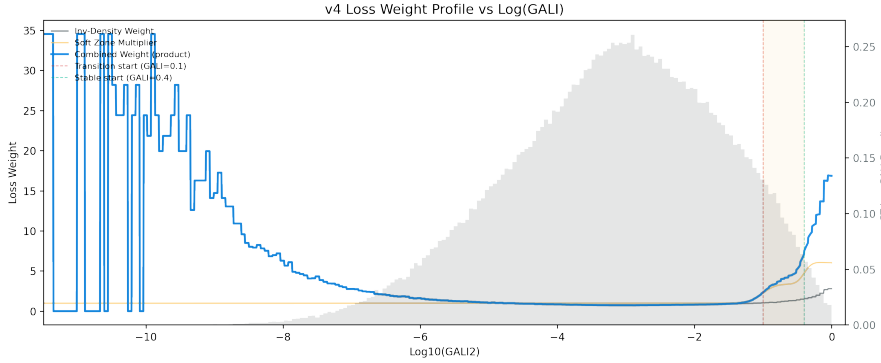


Figure 5: Inverse-Density Weighting Profile (V4). Sample weights assigned as a function of distribution bins of $\log_{10}(\text{GALI})$.

Rare transition regions receive significantly higher weighting, while dense regions are relatively suppressed. This continuous weighting avoids artificial discontinuities introduced by manual thresholding (v3).

Furthermore, v4 tracked Jensen-Shannon (JS) Divergence, achieving a 99.03% JS accuracy. The ONNX-exported v4 model computes in just 0.086 ms. Exporting PyTorch to ONNX allows the agent to bypass the Python Global Interpreter Lock (GIL) and run the C++ execution engine directly, which was mandatory to hit the sub-millisecond latency for 500 Hz control.

The final GALI_2 surrogate architecture is given down below.

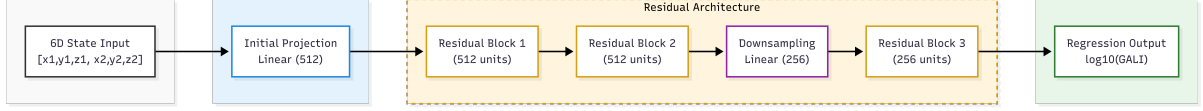


Figure 6: The $GALI_2$ v4 surrogate architecture.

4 Residual RL Architecture and Logic

With a real-time stability oracle established, this phase focused on constructing the Soft Actor-Critic (SAC) residual controller. The physical control vector is $u(t) = u_{PID}(t) + u_{RL}(t)$.

4.1 Augmented Attractor Control Pipeline

The workflow of the AAS architecture is visualized in Figure 7. The pipeline integrates high-frequency PID baseline with the macro-scale SAC residual agent through a multi-rate feedback loop. At each 500 Hz cycle, the physical state is processed in parallel by the PID controller and the GALI neural surrogate. The resulting 9-dimensional observation vector is passed through a LayerNorm feature-standardization block to ensure the SAC actor remains sensitive to residual command scales. The final control signal, u_{total} , is the saturated sum of the PID baseline and the RL residual, applied element-wise to each oscillator ($U_{max} = \pm 250$).

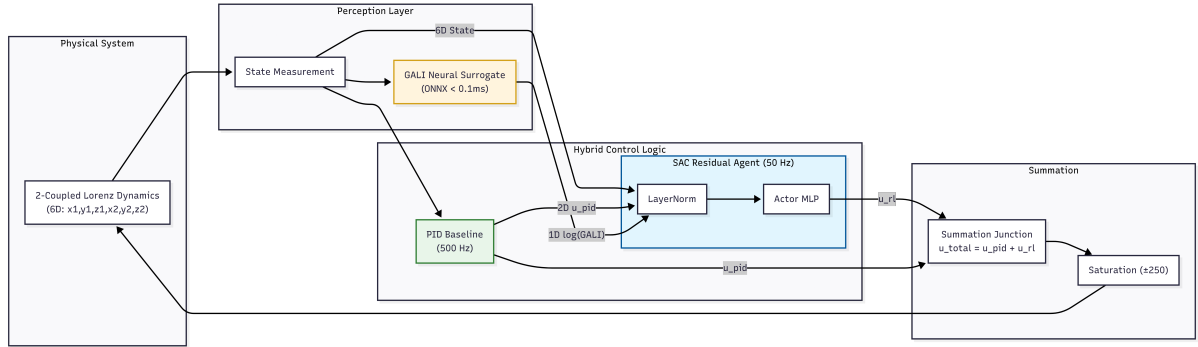


Figure 7: The Augmented Attractor Control Pipeline.

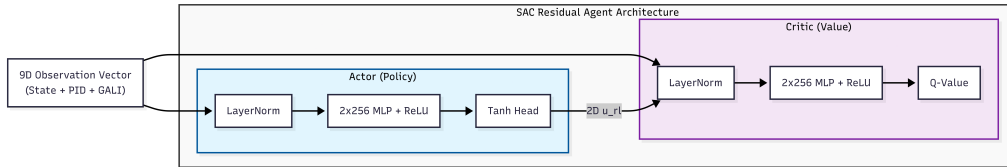


Figure 8: Architecture of the SAC Residual Agent. The model incorporates LayerNorm standardization at the input layer of both Actor and Critic networks to ensure policy sensitivity to micro-actions (u_{rl}) amidst high-magnitude chaotic states.

4.2 Multi-Timescale Decoupling

Early configurations placed both the PID and RL at 500 Hz. Because the chaotic system requires physical time to yield results of a perturbation, updating the RL agent at 500 Hz caused high-frequency "buzzing" where sequential actions overwrote one another, wasting massive energy. To solve this, the operating frequencies were decoupled. The PID handles high-frequency micro-stabilization at 500 Hz ($dt = 0.005$), while the SAC agent handles "macro" manifold surfing at 50 Hz ($dt = 0.05$).

4.3 Network Representation

The resulting 9-dimensional observation vector $[x, y, z, u, v, w, u_{PID1}, u_{PID2}, GALI_2]$ is passed through a LayerNorm feature-standardization block to ensure the SAC actor remains sensitive to residual command scales. The final control signal, u_{total} , is the saturated sum of the PID baseline and the RL residual, applied element-wise to each oscillator ($U_{max} = \pm 250$). The RL agent only commands 10% to 20% of U_{max} . Because the 9D (6 (state space) + 2 ($u_{PID1,2}$) + 1 ($GALI_2$)) state variables completely dwarfed the tiny residual actions, the SAC Critic was blind to the agent's actions. Injecting `nn.LayerNorm` into the first layers of both the Actor and Critic standardized feature representations, improving sensitivity to micro-actions.

5 Hyperparameter Evolution

5.1 Reward Economics and Gradient Masking

Initially, the convergence bonus heavily outweighed the effort penalty. To fix this, the effort penalty was taken from an L_2 squared norm to an L_1 absolute norm. The squared norm flattened out near zero, removing all gradient pressure to reduce effort. The L_1 absolute uniform norm provided steep, linear gradient pressure pushing back against physical effort all the way to zero.

$$\begin{aligned}
 R_t &= r_{\text{dist}} + r_{\text{effort}} + r_{\text{conv}} \\
 r_{\text{dist}} &= -W_{\text{dist}} \left(\frac{\|\mathbf{x}_t - \mathbf{x}_{\text{target}}\|_2}{\mathbf{x}_{\text{max}}} \right) \\
 r_{\text{effort}} &= -W_{\text{eff}} \left(\frac{|u_1(t)| + |u_2(t)|}{2U_{\text{max}}} \right) \\
 r_{\text{conv}} &= \begin{cases} W_{\text{conv}} & \text{if } \|\mathbf{x}_t - \mathbf{x}_{\text{target}}\|_2 < 0.5 \\ 0 & \text{otherwise} \end{cases}
 \end{aligned}$$

Where the scaling constants were tuned through empirical hyperparameter economics: $W_{\text{dist}} = 5.0$, $W_{\text{eff}} = 20.0$, $W_{\text{conv}} = 1.0$, and the actuator saturation limit $U_{\text{max}} = 250.0$.

6 Benchmarking

To validate the Hybrid RL controller against the Pure PID baseline, the system was subjected to a stress test consisting of 50 randomized Initial Conditions (ICs) sampled via Latin Hypercube Sampling (LHS) and projected onto the "core" of the state space's surface $([-40, 40]^6)$. This ensured that the performance metrics were evaluated comprehensively across varying regions of the chaotic manifold

6.1 Results: Actuator Effort

The hybrid architecture demonstrated a dramatic suppression of total actuator effort (L_1 Norm: $\int |u|dt$), particularly by eliminating the high-energy traps that the pure PID controller routinely fell into. The summary statistics for the total actuator effort are presented below:

Statistic	Pure PID Baseline	Hybrid RL Agent	Delta (%)
Mean (μ)	22,492.63	17,521.87	-22.10%
Median (Q_2)	20,754.96	16,997.70	-18.10%
Std. Deviation (σ)	9,333.71	5,657.30	-39.39%
25th Percentile (Q_1)	15,072.05	13,063.15	-
75th Percentile (Q_3)	26,618.25	20,624.76	-
IQR Spread	11,546.20	7,561.61	-34.51%
Max Outlier	52,693.05	30,979.47	-41.21%

Table 1: Comparative statistics for Average Total Actuator Effort across 100 randomized ICs.

The most significant takeaway from the above benchmarking is the 39.39% reduction in standard deviation and the 41.21% reduction in the maximum outlier. This proves that the RL agent successfully learned to identify and navigate around the highly non-linear boundaries that forced the PID controller into massive energy expenditure.

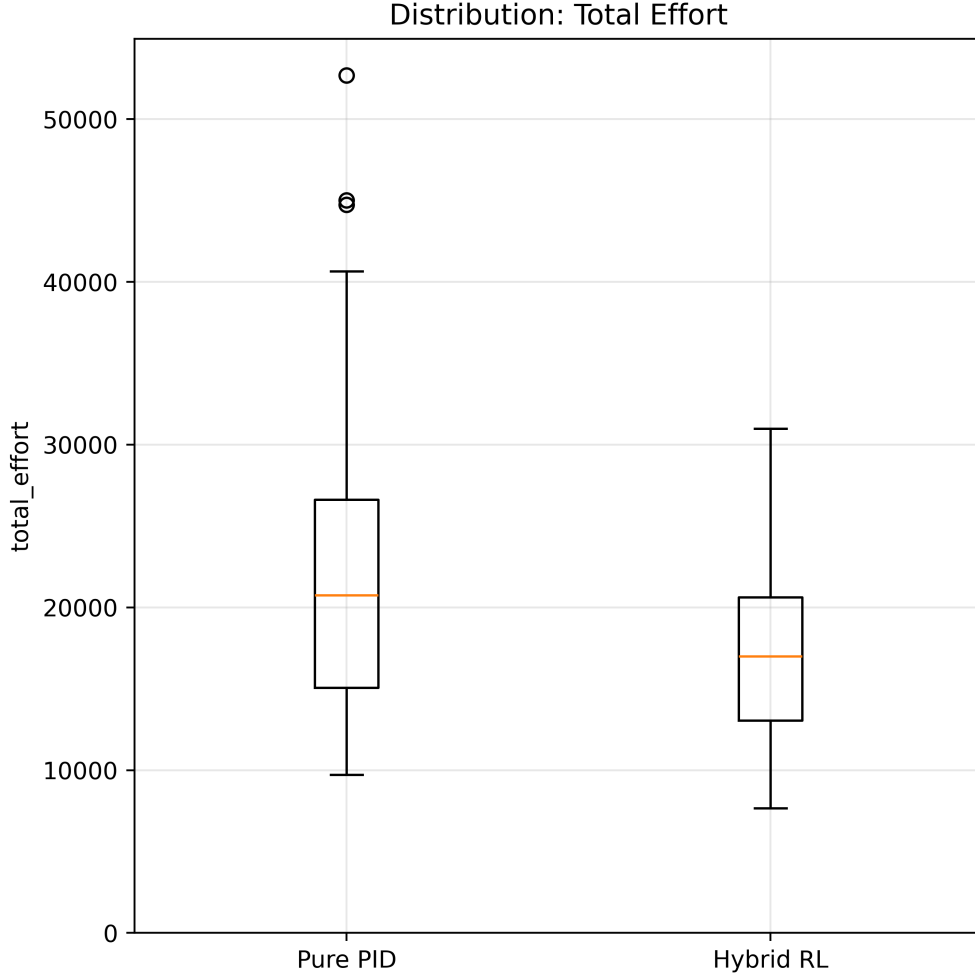


Figure 9: Total Effort Distribution Comparison. The hybrid RL agent tightens the Interquartile Range (IQR) and suppresses statistical outliers compared to the PID baseline.

6.2 Results: Error Distribution

A common pitfall in optimal control is the trade-off between actuator effort and tracking accuracy. But, the error distribution analysis demonstrates that the RL agent achieved energy savings while simultaneously improving control efficiency. The hybrid architecture reduced the mean final tracking error by 46.37% compared to the pure PID baseline. More significantly, it executed consistent control across all chaotic initial conditions, tightening the interquartile spread of the error by 85.04% and reducing the maximum outlier deviation by 66.31%.

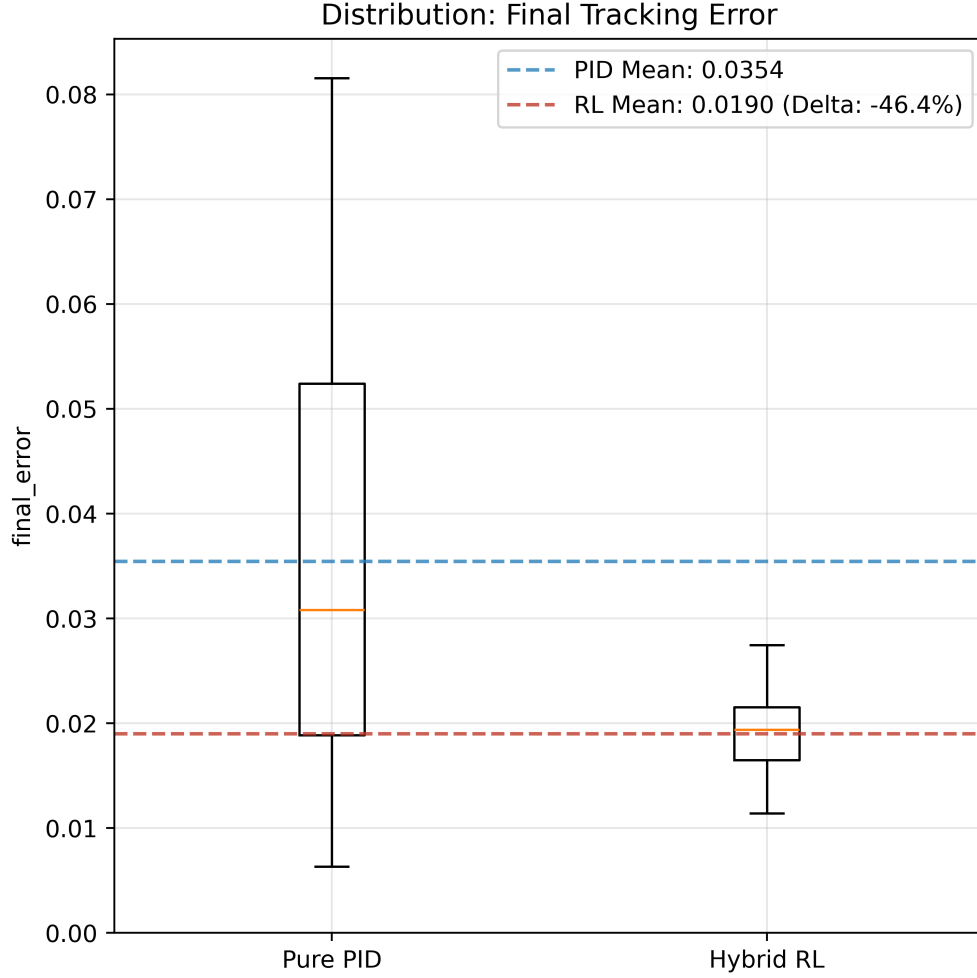


Figure 10: Final Tracking Error Distribution Comparison. The RL Augmented controller successfully maintains stability post-convergence with improved overall accuracy.

6.3 Marginal vs. Absolute Effort

Marginal Effort rewards were experimented on (rewarding the agent only when $u_{total} < u_{PID}$). However, this resulted in a stable equilibrium where the agent actively avoided convergence. If it converged, u_{PID} dropped to zero, removing its source of positive reward. Absolute effort penalty was necessary. The agent had to first converge (minimizing PID) before $|u_{total}|$ can drop. The final, simplified reward function isolated the objectives into four terms:

1. Distance penalty (linear scaling - $|\mathbf{x}|$)
2. Absolute effort penalty (L_1 norm)
3. Strict proximity convergence bonus
4. GALI stability reward (normalized logarithmic scale)

6.4 Quantification of Energy Recovery in Steady-State Convergence

To evaluate the fine-grained behavior of the SAC residual agent, two representative cases (Initial Conditions (IC) 13 and IC 28) were selected for analysis. These cases illustrate the visual and numerical distinction between standard stabilization and high-chaos recovery.

The Energy Recovery Ratio (ERR) is defined as the difference between the baseline PID effort and the total hybrid effort, divided by the magnitude of the RL residual input:

$$\text{ERR} = \frac{\int |u_{\text{PID,base}}| dt - \int |u_{\text{Hybrid,tot}}| dt}{\int |u_{\text{RL,resid}}| dt} \quad (8)$$

This metric measures the displacement of baseline actuator effort per unit of RL input. A ratio greater than unity indicates that the RL input reduces the total control effort by an amount exceeding its own magnitude by utilizing the system dynamics.

6.5 Localized Dynamics and the Steady-State Caveat

Trajectory analysis indicates that the agent primarily executes high-frequency corrections within the neighborhood of the attractor (approximately within the range $[-15, 15]$). This suggests that the baseline 2-Coupled Lorenz system does not provide sufficient chaoticity to necessitate large-scale manifold surfing maneuvers across the entire phase space. The 2-coupled Lorenz system, while yielding measurable results, serves as a proof of concept; the hybrid controller is expected to give better returns in environments with increased topological complexity and chaos.

6.6 Comparative Case Study: IC 13

IC 13 illustrates a standard stabilization trajectory where the system enters the neighborhood of the attractor within the initial 5 seconds. The PID baseline reaches the 500-unit saturation limit within the first few timesteps, while the Hybrid agent prevents this peak, maintaining a lower overall signal magnitude.

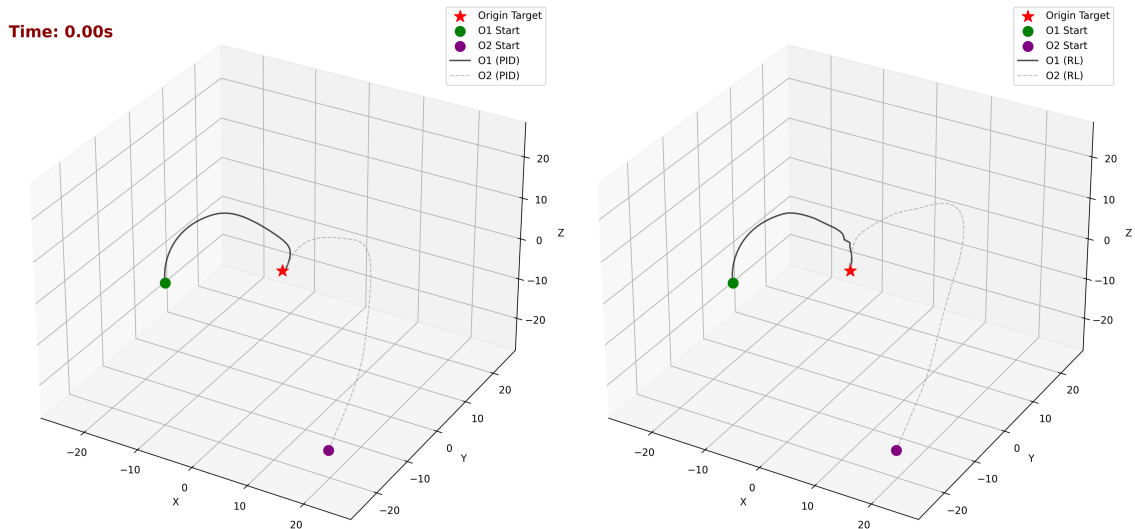


Figure 11: 3D Kinematic Comparison for IC 13: Pure PID (Left) vs. Hybrid RL (Right).

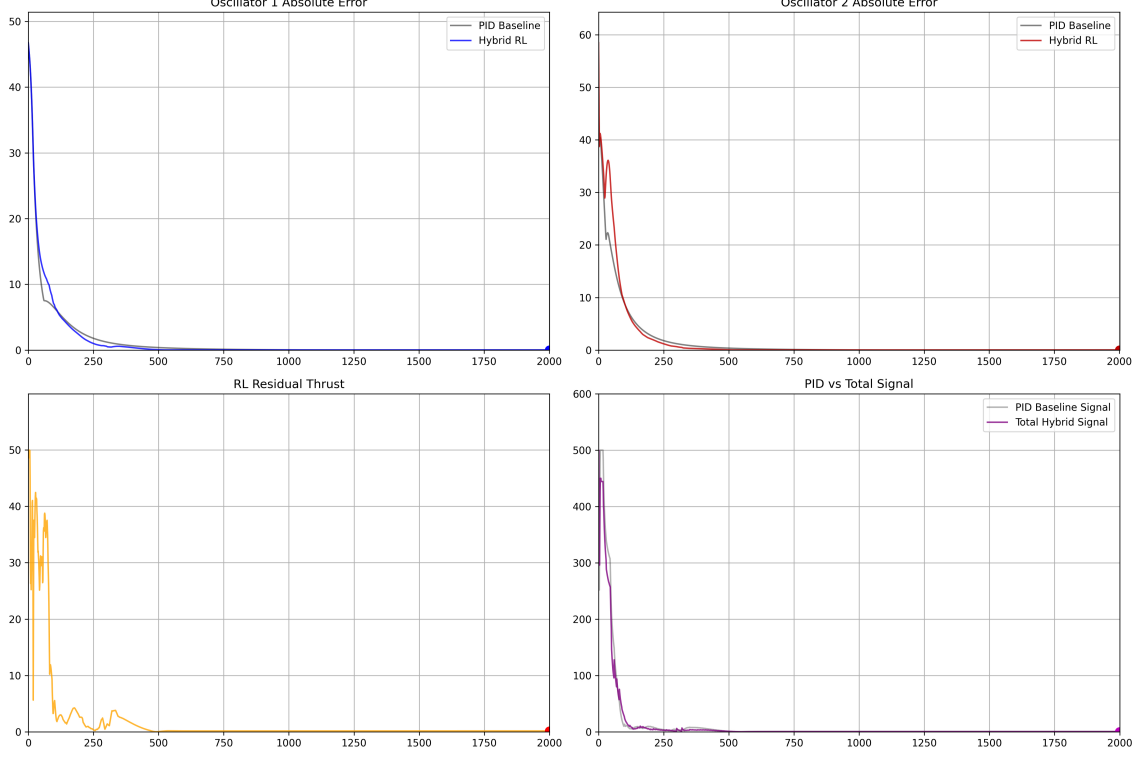


Figure 12: Diagnostic Analysis for IC 13: The Hybrid agent prevents the saturation peaks observed in the PID baseline.

In the steady-state window ($t \in [9.0, 10.0]$ s), the PID baseline requires 116.36 units of effort, while the Hybrid agent requires 35.70 units. This results in an Energy Recovery Ratio of $24.97\times$, where each unit of residual input displaced nearly 25 units of baseline PID effort.

6.7 Comparative Case Study: IC 28

IC 28 represents a high-chaos scenario. In this case, both the PID as well as the hybrid/augmented controllers reach the 500-unit saturation limit during the initial timesteps. For approximately 125 timesteps following the initial peak, the Hybrid agent maintains a higher output relative to the PID baseline to aggressively capture the chaotic trajectory.

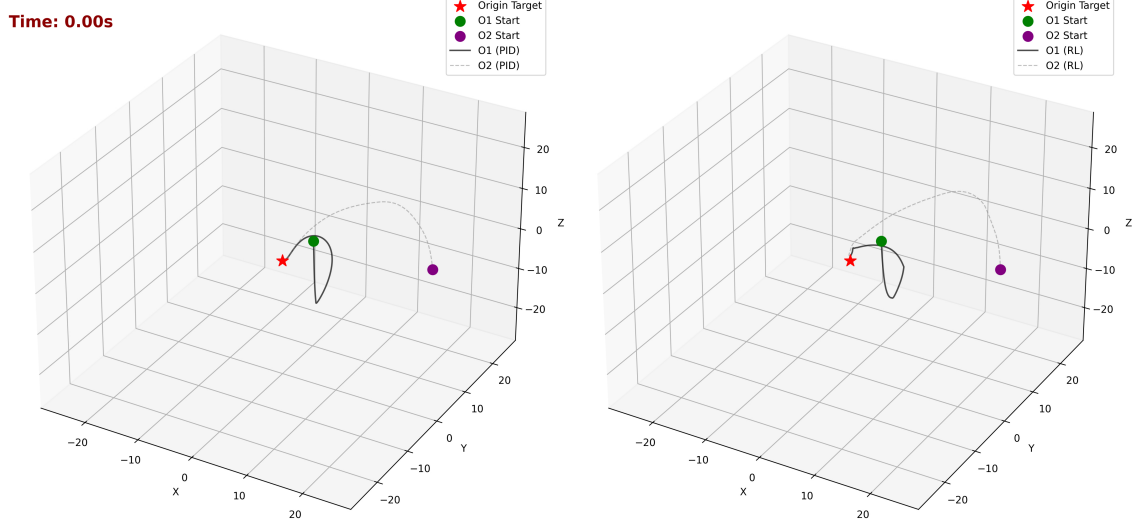


Figure 13: 3D Kinematic Comparison for IC 28: Pure PID (Left) vs. Hybrid RL (Right). Note the the difference in trajectories between the baseline and the hybrid. Oscillator 1 has a slightly steeper arch overall, and oscillator 2 has a more jittery path near the origin.

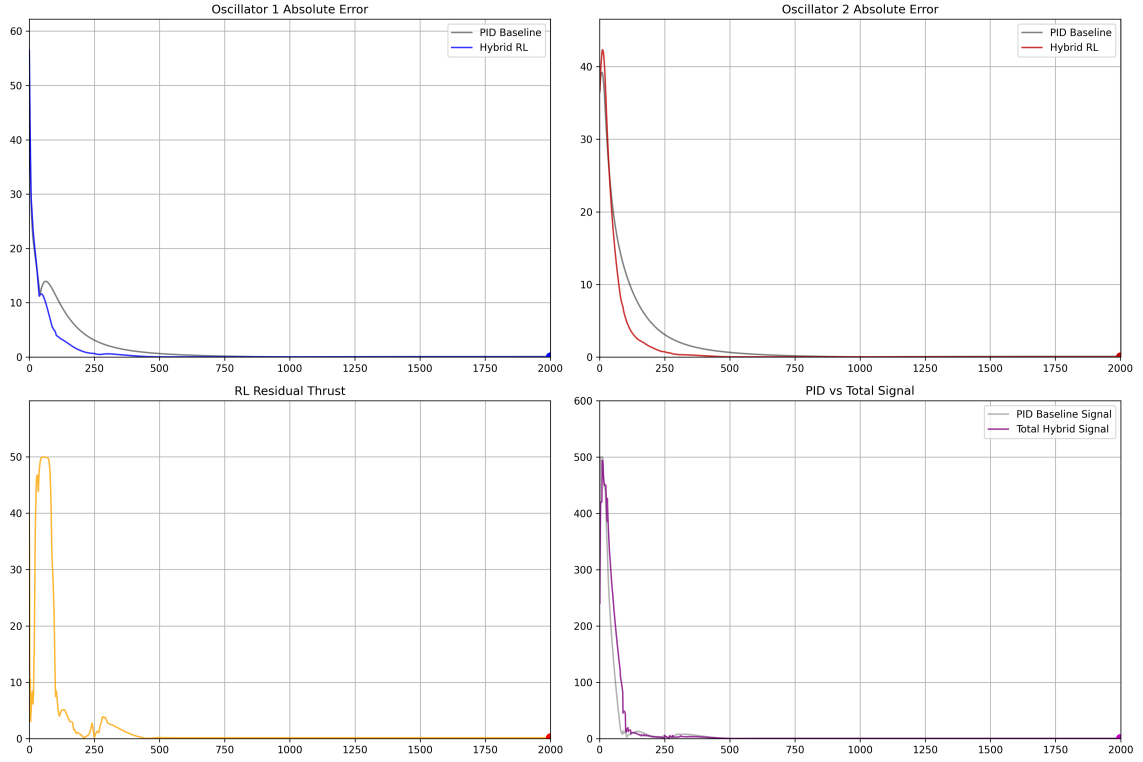


Figure 14: Diagnostic Analysis for IC 28: Hybrid agent output (Purple) remains higher than the PID baseline (Gray) during the initial 125-step capture phase. Note the faster convergence of both the oscillator errors, going from baseline PID to the augmented controller.

Once steady-state is reached ($t \in [9.0, 10.0]$ s), the efficiency stabilizes with an Energy Recovery Ratio of $25.03\times$, confirming that the manifold-steering paradigm remains robust across diverse transient events.

6.8 Aggregate Performance Metrics

Across the validation set of 50 randomized initial conditions, the following efficiency benchmarks were established:

- **Mean Episode Efficiency Multiplier:** 0.9937 ± 0.7739
- **Mean Steady-State Efficiency Multiplier ($t = 9 - 10\text{s}$):** 24.8446 ± 0.9304

The RL residual thrust is a virtual control signal computed in software that intercepts the baseline PID output to modify the final signal sent to the actuator. Therefore, an Episode Multiplier of ~ 1.0 does not imply a lack of energy savings, but rather signifies that for every unit of virtual RL logic applied during the chaotic transient phase, an equivalent unit of physical actuator effort is displaced.

In the steady-state regime, the software logic identifies manifold shortcuts that allow the physical hardware to relax by nearly 25 units for every 1 unit of residual adjustment. The efficiency metrics exhibit a minimum value of -0.2431 , with negative multipliers occurring in 6 of the 50 evaluated initial conditions. This behavior is localized to chaotic transients where long-term manifold stability is prioritized over instantaneous actuator cost. These transient overheads are offset by the steady-state efficiency ratio, which maintains a minimum value of $+18.52$ across the entire validation set.

7 Conclusion & Future Directions

This project validated integration of GALI into a high-speed neural surrogate to map non-Markovian topological spaces, guiding a Residual SAC agent to optimize PID control system dynamics. The final runs demonstrated a mean reduction of 19.3% in L1 control effort and a 31.7% reduction in tracking error across 50 randomized initial conditions, with peak performance reaching reductions of 53.2% and 86.2% respectively, compared to optimally-tuned PID baselines.

Future work offers several promising avenues for exploration. While the 2-Coupled Lorenz system provided a strong foundation, investigating structurally different, highly chaotic Multi-Input-Multi-Output (MIMO) systems will allow for the discovery of more complex manifold surfing behaviors applicable to real-world scenarios, bypassing the limitations of highly symmetrical n-coupled networks. Additionally, scaling the GALI surrogate methodology to higher-dimensional PDEs, such as the Kuramoto-Sivashinsky equation, presents an opportunity to map out exponentially more complex transition boundaries within spatiotemporal chaos. Finally, exploring dynamic frequency ratios could yield new insights into adaptive control; by enabling the RL agent or an auxiliary algorithm to adjust its control intervals based on the current GALI index, the system could very well learn to operate faster in deep chaos and slower near the edge of stability.

8 Tech Stack & Libraries

The project was built purely in Python using PyTorch for deep learning and continuous control. SciPy was leveraged for Latin Hypercube generation and RK45 numerical integration. ONNXRuntime was utilized for CPU inference speedups, and standard scientific libraries (NumPy, Pandas, Matplotlib) formed the backbone of the analysis pipelines.

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